

Mostafa Bamdad

Scientific Machine Learning Engineer

Research Assistant, Bauhaus-Universität Weimar
Weimar, Germany

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Profile

Research Assistant at Bauhaus-Universität Weimar with a background in computational mechanics, scientific machine learning, and numerical simulation. Experienced in developing data-driven models for complex physical systems and translating mathematical and engineering concepts into reliable computational solutions. Strong analytical ability, independent research experience, and a proven record of interdisciplinary collaboration and scientific publications.

Experiences

Research Assistant in Computational Mechanics and Scientific Machine Learning Since 06/2022
Bauhaus-Universität Weimar

- Develop machine-learning models and neural operators for nonlinear, time-dependent, and multiphysics systems.
- Build reproducible workflows for data preparation, model training, validation, benchmarking, and error analysis.
- Implement scientific computing solutions using Python, PyTorch, TensorFlow, JAX, MATLAB, Abaqus, and FEniCS.
- Compare data-driven approaches with numerical reference methods and evaluate model accuracy, stability, and generalization.
- Collaborate across machine learning, mechanics, applied mathematics, and engineering research.
- Communicate results through technical documentation, open-source implementations, scientific publications, and presentations.

Education

Since 06/2022 Computational Mechanics, Bauhaus-Universität Weimar, Germany
09/2016 – 06/2019 M.Sc. Mechanical Engineering, University of Kashan, Iran
09/2012 – 09/2016 B.Sc. Mechanical Engineering, University of Guilan, Iran

Technical Expertise

Machine Learning	PyTorch, TensorFlow, scikit-learn, JAX, neural networks, model evaluation, optimization, and scientific machine learning
Programming and Data	Python, NumPy, pandas, SciPy, Matplotlib, MATLAB, data preparation, analysis, and visualization
Generative AI	Professional prompt engineering, LLM integration, structured prompting, context design, and output evaluation
Scientific Computing	PDE modelling, numerical methods, FEM/FEA, Abaqus, FEniCS, and multiphysics simulation
Development Tools	Git, GitHub, Linux, Jupyter, modular implementation, debugging, and technical documentation

Languages

German: C1 **English: Fluent** **Persian: Native**

Selected Projects

- **HAMNO** – Hierarchical adaptive multi-scale neural operator for nonlinear dynamical systems with non-periodic boundary conditions and stable long-horizon prediction.
- **PENCO** – Physics–energy–numerics-consistent neural operator for accurate and stable three-dimensional phase-field simulation.
- **Scientific Machine Learning for Structural Systems** – JAX-based models connecting machine learning, PDE modelling, and numerical mechanics. [DOI](#)

Research Output and Conferences

Selected Publications

- Developer of **HAMNO** (Hierarchical Adaptive Multi-Scale Neural Operator). Preprint, 2026.
- Developer of **PENCO** (Physics–Energy–Numerics–Consistent Operator) for 3D phase-field modeling, *CMAME*, 2026.
- Scientific Machine Learning for PDE-Based Structural Systems. *Neurocomputing*, 2024. [DOI](#)
- Multiphysics Modeling of Magneto-Electro-Elastic Sandwich Structures. *Advances in Nano Research*, 2023. [DOI](#)

Conferences and Workshops

- **Advancing Scientific Machine Learning in Industry**, Technical University of Munich Institute for Advanced Study, Germany, October 15–16, 2024. Presented a poster on machine learning for PDE-based computational mechanics.
- **Machine Learning and PDEs – MLPDES25 Workshop**, FAU Research Center for Mathematics of Data, Erlangen, Germany, April 28–30, 2025.

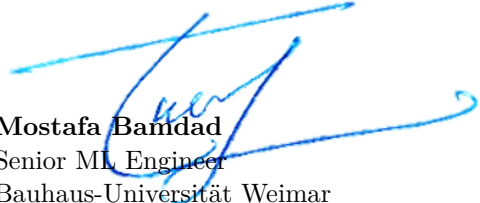
Awards and Training

Award	DAAD Scholarship, German Academic Exchange Service, 2023 – Present
Training	Python for Computational Mechanics; Completed Data Science Bootcamp; Advanced Static and Dynamic Simulation

Reference

Professor Timon Rabczuk

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Mostafa Bamdad
Senior ML Engineer
Bauhaus-Universität Weimar
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